

Total number of pages: 1
Degree: B.Tech.

Roll no.:
Semester: 3rd

Course work
MID-SEMESTER EXAMINATION, September 2025

Course code: CMMTC302

Course title: Applied Linear Algebra

Time: 1 hour 30 min.

Maximum Marks. 20

Note: Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed and mentioned in the answer. Here, we follow the same notations used in class.

Q. No.	Question	Marks	CO
1a	Are 48 and 23 in the same residue class modulo 8?	2	CO1
1b	Prove that the set $G := \left\{ \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \mid x \in \mathbb{R} \right\}$ is an abelian group, here $\mathbb{R}$ denotes the set of real numbers.	2	CO1
2a	Let V be a vector space of all 2×2 matrices over the field of real numbers. Let W_1 be the set of matrices of the following form $\begin{pmatrix} x & -x \\ y & z \end{pmatrix}$ and let W_2 be the set of matrices of the form $\begin{pmatrix} a & b \\ -a & c \end{pmatrix}.$ Prove that W_1 and W_2 are subspaces of V .	2	CO1
2b	Find the dimensions of the subspace $W_1 + W_2$, where W_1 and W_2 are defined in Question 2 (a).	2	CO1
3a	Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the map defined by $T(x_1, x_2) = (x_2, x_1)$. Is T a linear map? If so, find its null and range space.	2	CG2
3b	Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear map defined by $T(x_1, x_2) = (x_1 - x_2, x_1)$. Is T invertible? If so, find its inverse.	2	CO2
4a	Let $T : V \rightarrow W$ be a linear transformation and let $\{T(v_1), T(v_2), \dots, T(v_n)\}$ be linearly independent in the range of T i.e. in $R(T)$. Prove that $\{v_1, v_2, \dots, v_n\} \subset V$ is linearly independent.	2	CO2
4b	Does there exist a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ whose range space is generated by the vectors $(1, 2, 0)$, $(0, 1, 1)$, and $(1, 3, 1)$? If so, how many such transformations exist?	2	CO2
5a	Let V be a finite dimensional vector space over the field F . Let $T : V \rightarrow V$ be a linear map such that the null space of T and the range space of T are equal (i.e., $N(T) = R(T)$). Prove that $\dim_F(V)$ is always even.	2	CO2
5b	Find a linear map from $\mathbb{R}^2$ to $\mathbb{R}^2$ such that its range and null space are equal.	2	CO2